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**Course Code:** DSA0604

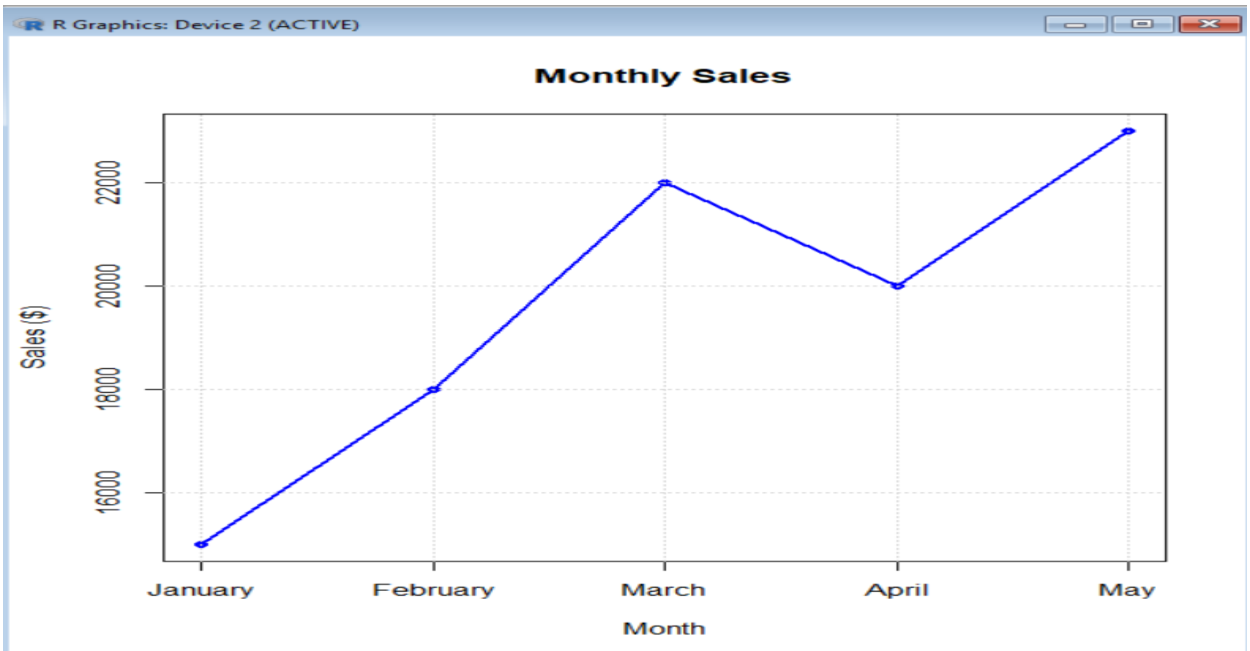
# Lab Experiments

## Set 1

## Program 1: Line Chart for Monthly Sales

```
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)
plot(sales,
    type = "o",
    xaxt = "n",
    xlab = "Month",
    ylab = "Sales ($)",
    main = "Monthly Sales",
    col = "blue",
    lwd = 2,
    pch = 16)
axis(1, at = 1:5, labels = month)
grid()
```

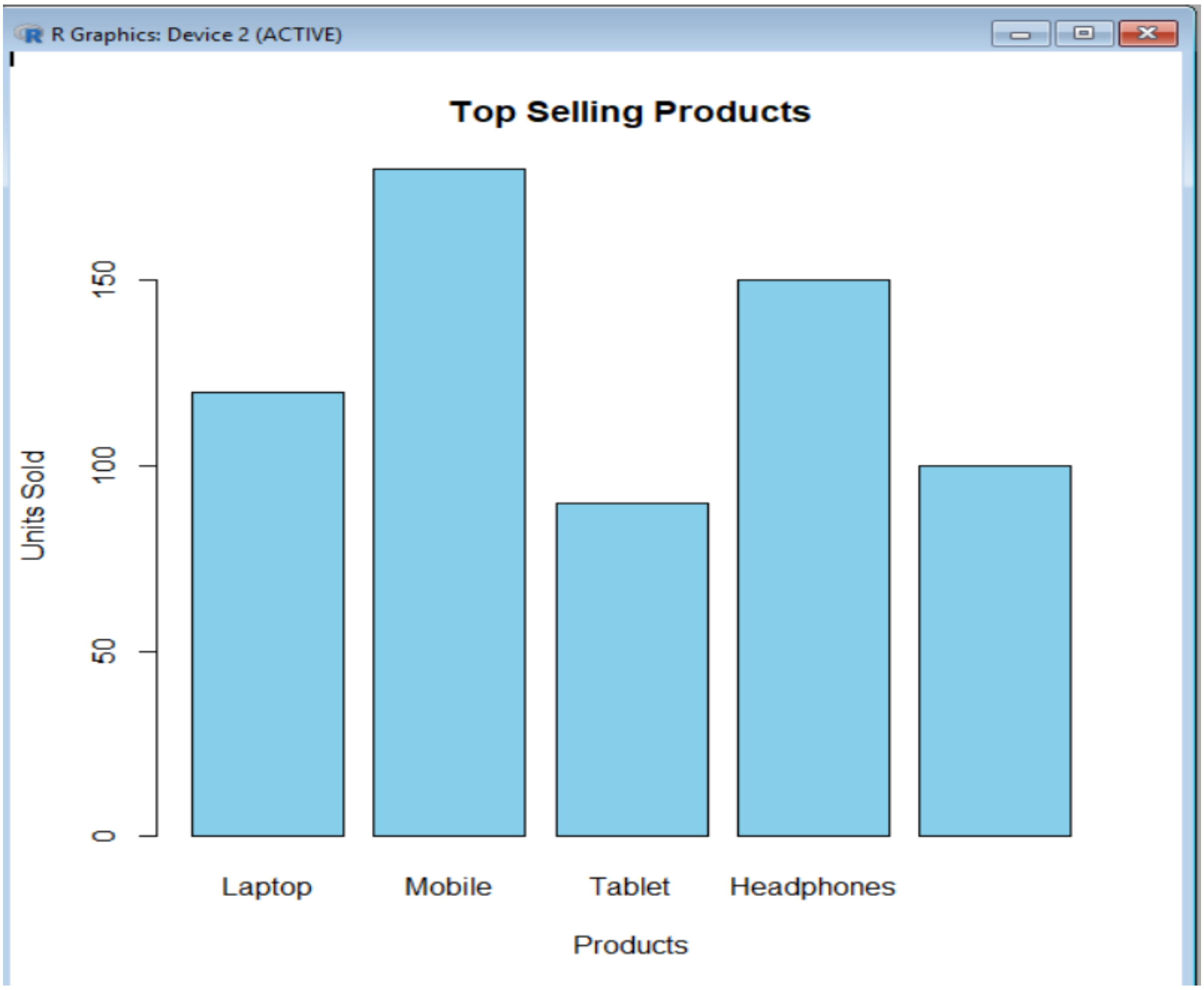
## Output:



## Program 2: Bar Chart for Top-Selling Products

```
products <- c("Laptop", "Mobile", "Tablet", "Headphones", "Smartwatch")
sales <- c(120, 180, 90, 150, 100)
barplot(sales,
        names.arg = products,
        col = "skyblue",
        main = "Top Selling Products",
        xlab = "Products",
        ylab = "Units Sold",
        border = "black")
```

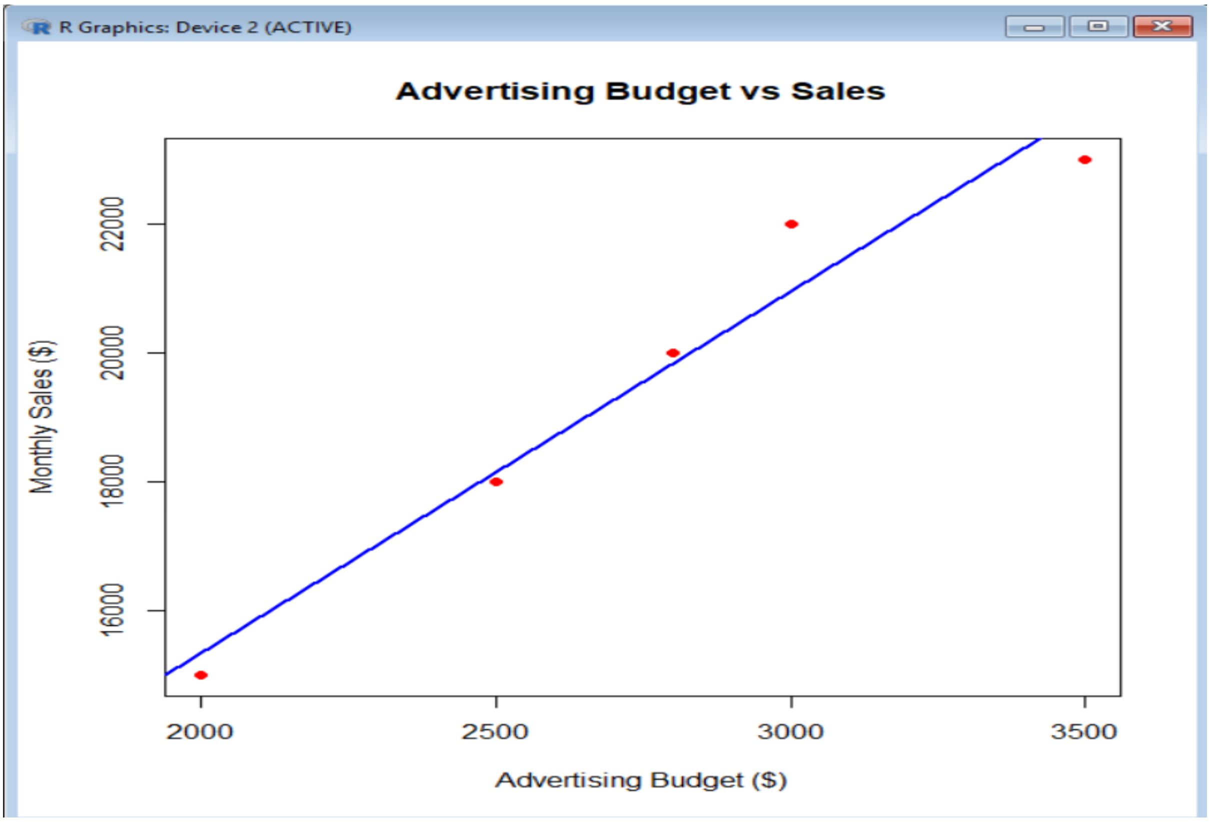
## Output:



## Program 3: Scatter Plot (Advertising Budget vs Monthly Sales)

```
advertising <- c(2000, 2500, 3000, 2800, 3500)
sales <- c(15000, 18000, 22000, 20000, 23000)
plot(advertising,
     sales,
     main = "Advertising Budget vs Sales",
     xlab = "Advertising Budget ($)",
     ylab = "Monthly Sales ($)",
     pch = 19,
     col = "red")
abline(lm(sales ~ advertising), col = "blue", lwd = 2)
```

## Output:

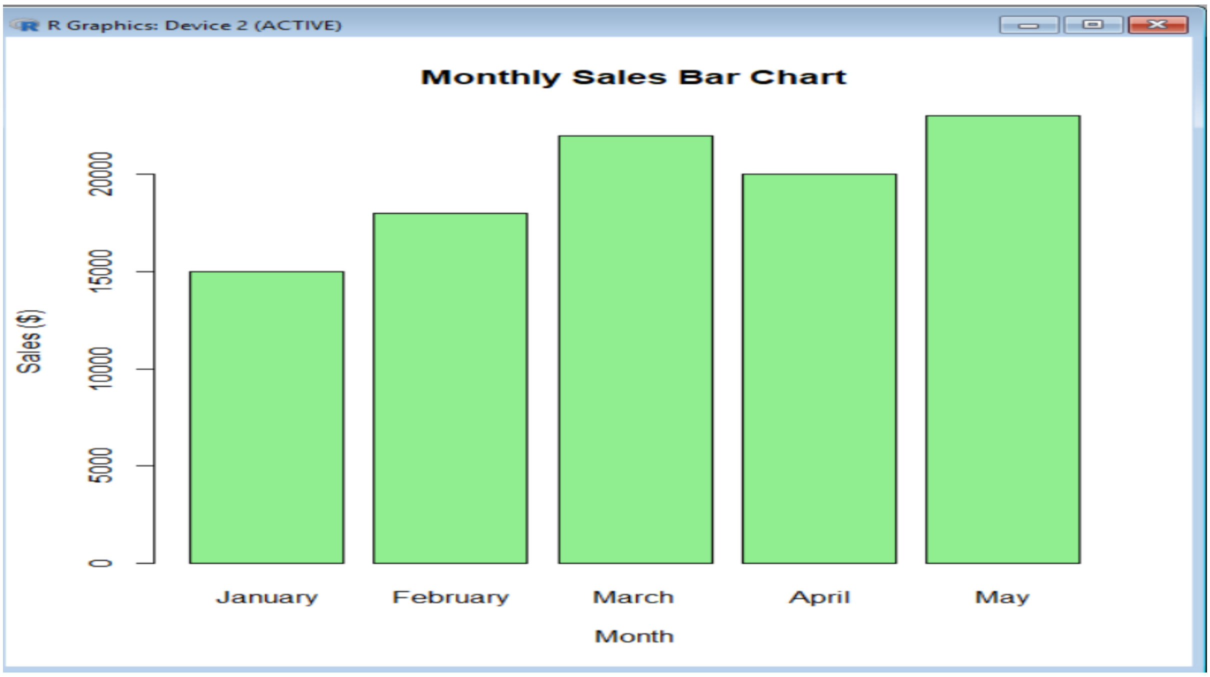


# Program 4: Interactive Dashboard (Line Chart + Bar Chart)

```
install.packages("manipulate")
library(manipulate)
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)
manipulate(
{
par(mfrow = c(1,2))
plot(sales,
    type = "o",
    xaxt = "n",
    xlab = "Month",
    ylab = "Sales ($)",
    main = "Monthly Sales",
    col = "blue",
    lwd = 2,
    pch = 16)
axis(1, at = 1:5, labels = month)
barplot(sales,
      names.arg = month,
      col = "lightgreen",
      main = "Monthly Sales Bar Chart",
      xlab = "Month",
      ylab = "Sales ($)")
},
```

dummy = slider(1,5,initial = 1))

Output:

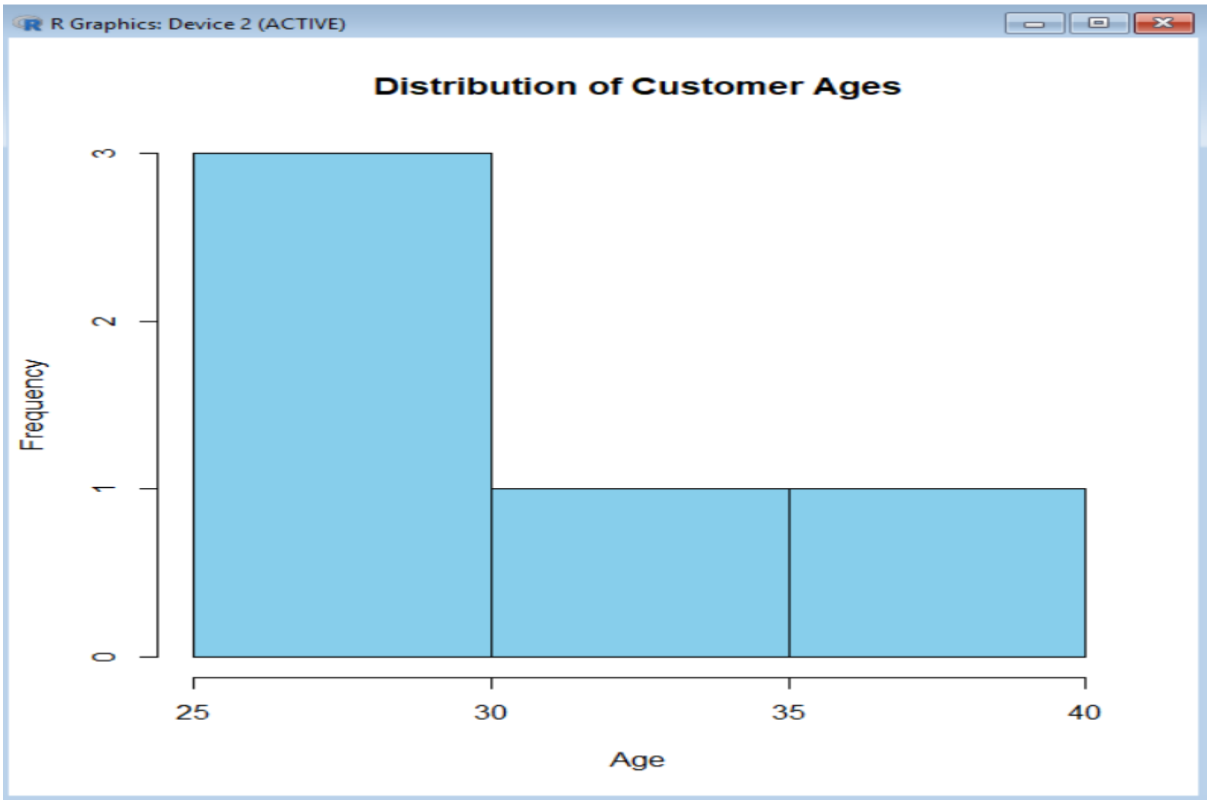



## Set 2

## Program 1: Histogram of Customer Ages

```
age <- c(25, 30, 35, 28, 40)
hist(age,
     main = "Distribution of Customer Ages",
     xlab = "Age",
     ylab = "Frequency",
     col = "skyblue",
     border = "black")
```

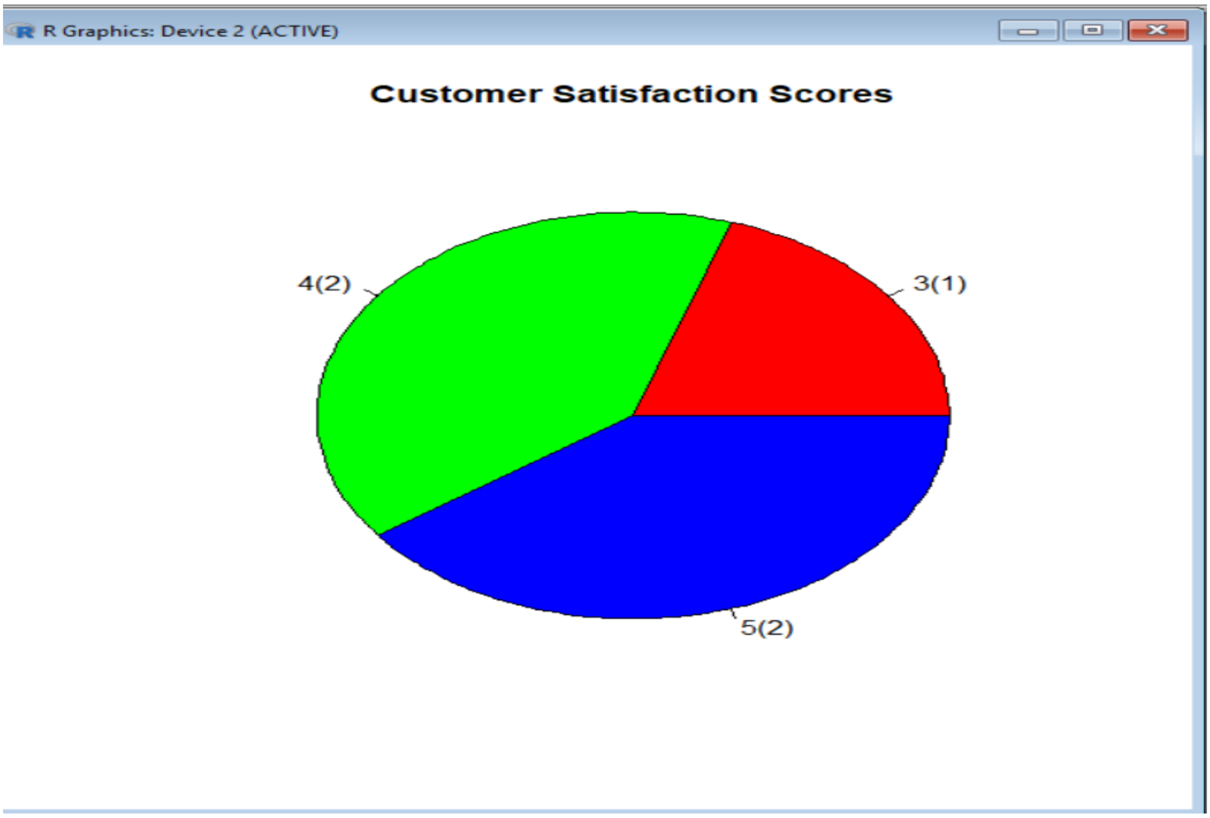
# Output:




# Program 2: Pie Chart of Customer Satisfaction Scores

```
customer_id <- c(1,2,3,4,5)

age <- c(25,30,35,28,40)

satisfaction <- c(4,5,3,4,5)

data <- data.frame(customer_id, age, satisfaction)

score_count <- table(data$satisfaction)

pie(score_count,

    main="Customer Satisfaction Scores",

    labels=paste(names(score_count), "(", score_count, ")", sep=""),

    col=rainbow(length(score_count)))
```

## Output:




## 3. Stacked Bar Chart of Satisfaction Scores by Age Group

```
customer_id <- c(1,2,3,4,5)

age <- c(25,30,35,28,40)

satisfaction <- c(4,5,3,4,5)

data <- data.frame(customer_id, age, satisfaction)

data$AgeGroup <- cut(data$age,

                 breaks=c(20,30,40,50),

                 labels=c("21-30","31-40","41-50"),

                 include.lowest=TRUE)

table_data <- table(data$AgeGroup, data$satisfaction)
```

```
barplot(table_data,
        main="Satisfaction Scores by Age Group",
        xlab="Age Group",
        ylab="Number of Customers",
        col=rainbow(ncol(table_data)),
        legend=colnames(table_data))
```

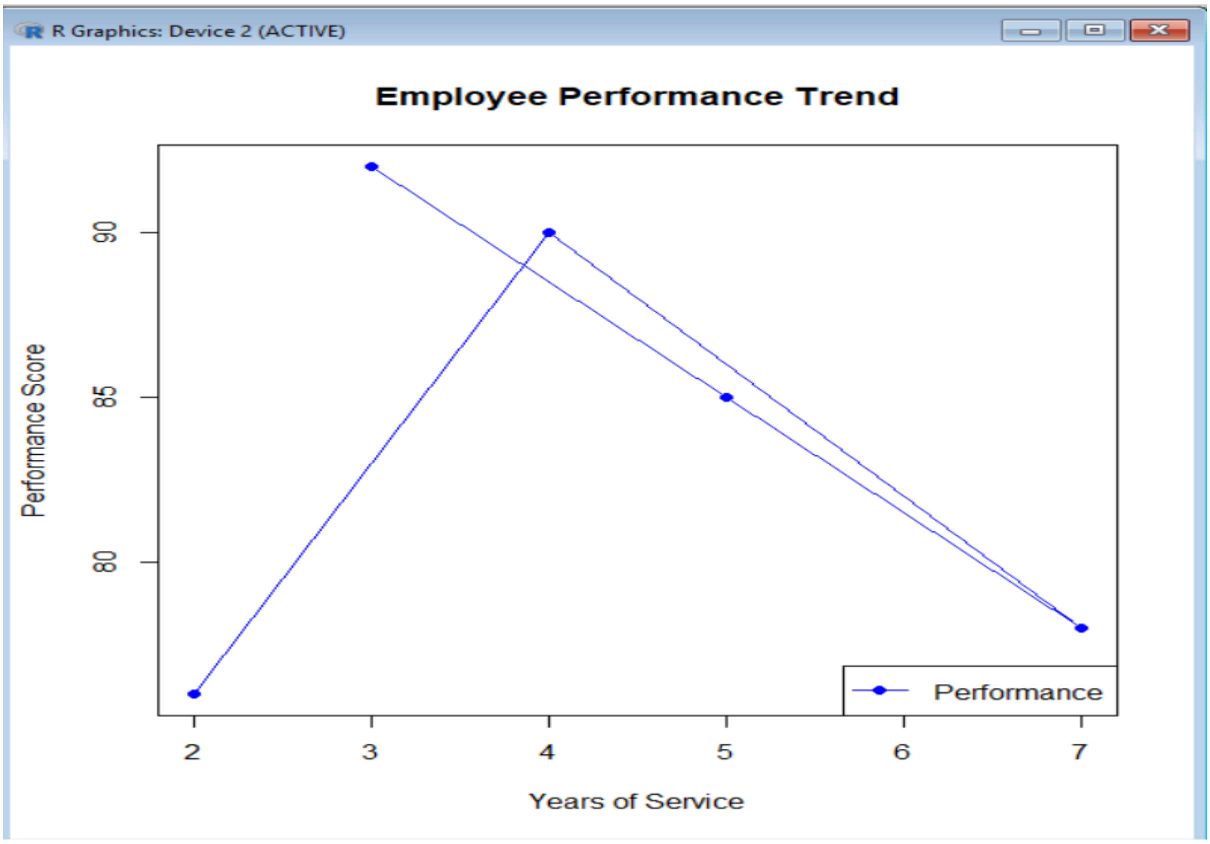
# Output:

# Set 3

# Program 1: Line Chart – Performance Trend of Employees

```
EmployeeID <- c(1,2,3,4,5)

Department <- c("Sales","HR","Marketing","Sales","HR")

YearsService <- c(5,3,7,4,2)

PerformanceScore <- c(85,92,78,90,76)

data <- data.frame(EmployeeID, Department, YearsService, PerformanceScore)

plot(data$YearsService,
     data$PerformanceScore,
     type="o",
     col="blue",
     pch=16,
     xlab="Years of Service",
     ylab="Performance Score",
     main="Employee Performance Trend")

legend("bottomright",
       legend="Performance",
       col="blue",
       lty=1,
       pch=16)
```

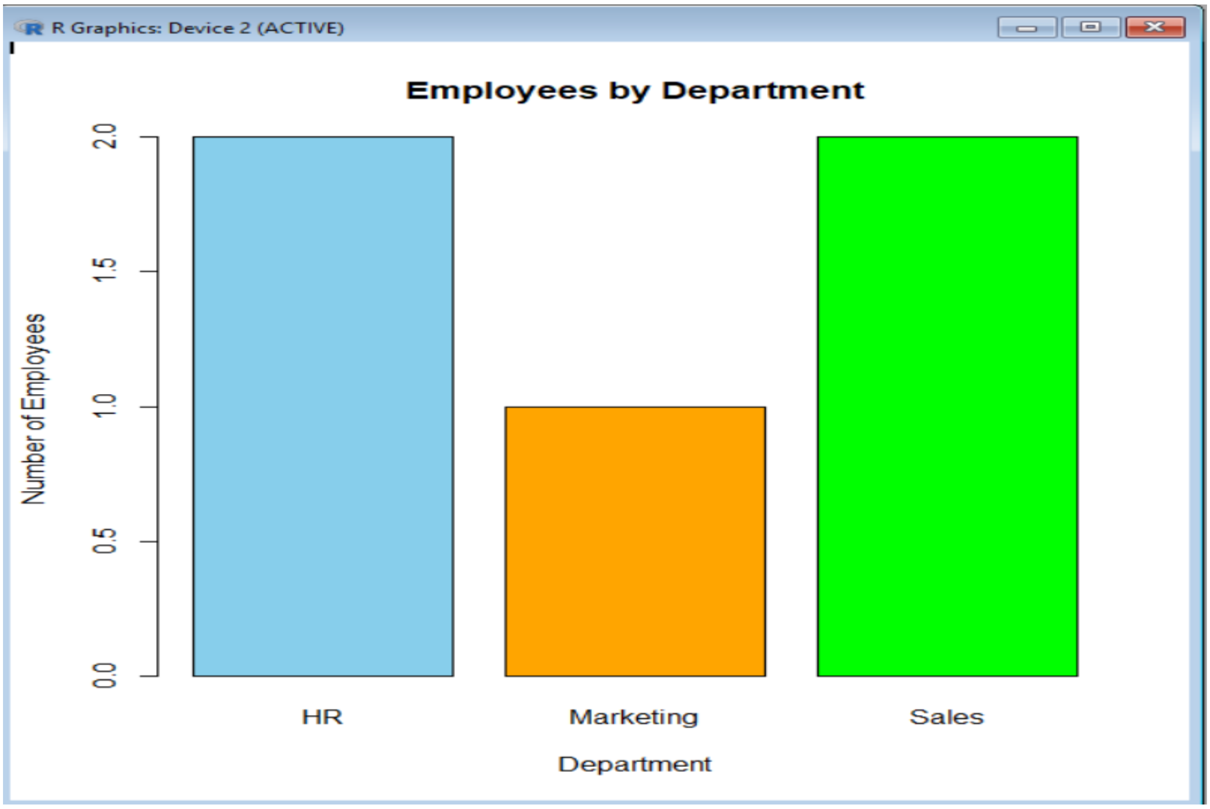
## Output:




# Program 2: Bar Chart – Employees by Department

```
EmployeeID <- c(1,2,3,4,5)

Department <- c("Sales","HR","Marketing","Sales","HR")

YearsService <- c(5,3,7,4,2)

PerformanceScore <- c(85,92,78,90,76)

data <- data.frame(EmployeeID, Department, YearsService, PerformanceScore)

dept_count <- table(data$Department)


barplot(dept_count,

        col=c("skyblue","orange","green"),

        main="Employees by Department",

        xlab="Department",

 ylab="Number of Employees")
```

## Output:

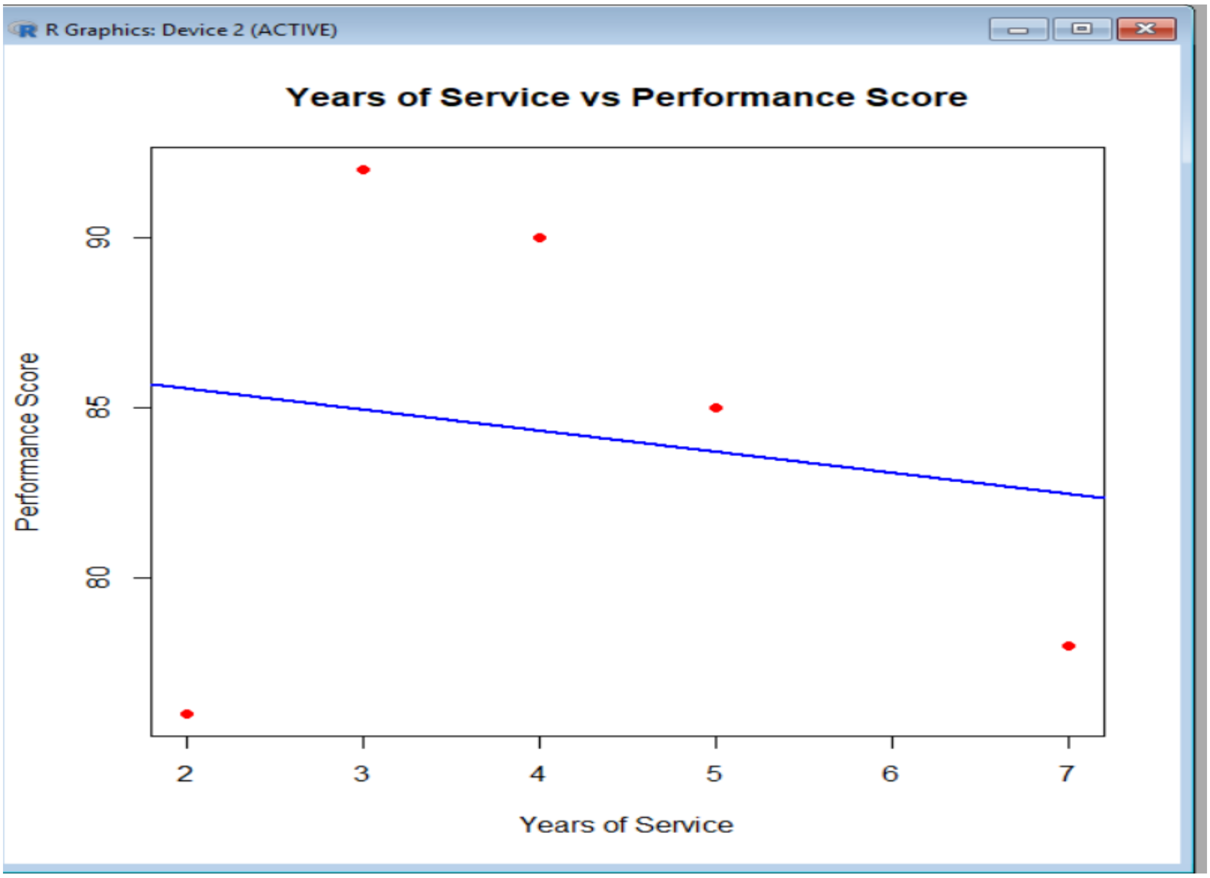



## Program 3: Scatter Plot – Years of Service vs Performance Score

```
EmployeeID <- c(1,2,3,4,5)
Department <- c("Sales","HR","Marketing","Sales","HR")
YearsService <- c(5,3,7,4,2)
PerformanceScore <- c(85,92,78,90,76)
data <- data.frame(EmployeeID, Department, YearsService, PerformanceScore)
plot(data$YearsService,
    data$PerformanceScore,
    pch=19,
    col="red",
    xlab="Years of Service",
```

```
    ylab="Performance Score",
    main="Years of Service vs Performance Score")
abline(lm(PerformanceScore ~ YearsService, data=data),
       col="blue",
       lwd=2)
```

# Output:

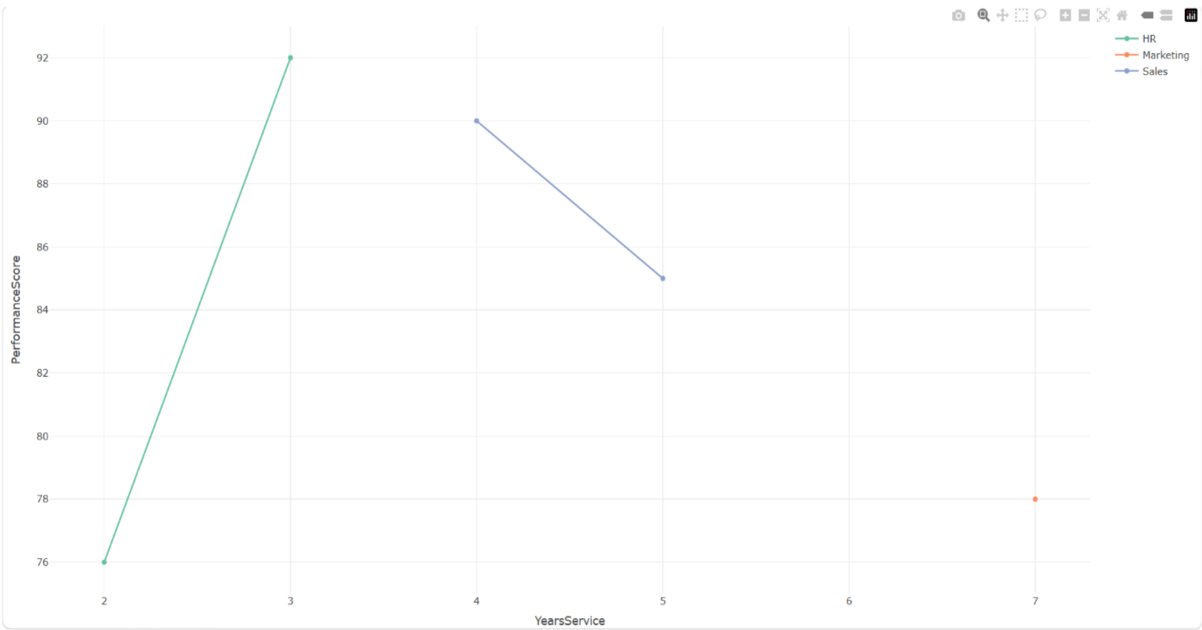



# Program 4: Interactive Dashboard (Using plotly)

```
library(plotly)
EmployeeID <- c(1,2,3,4,5)
Department <- c("Sales","HR","Marketing","Sales","HR")
YearsService <- c(5,3,7,4,2)
PerformanceScore <- c(85,92,78,90,76)
```

```
data <- data.frame(EmployeeID, Department, YearsService, PerformanceScore)
fig <- plot_ly(data,
            x=~YearsService,
            y=~PerformanceScore,
            type="scatter",
            mode="lines+markers",
            color=~Department)
fig
```

# Output:



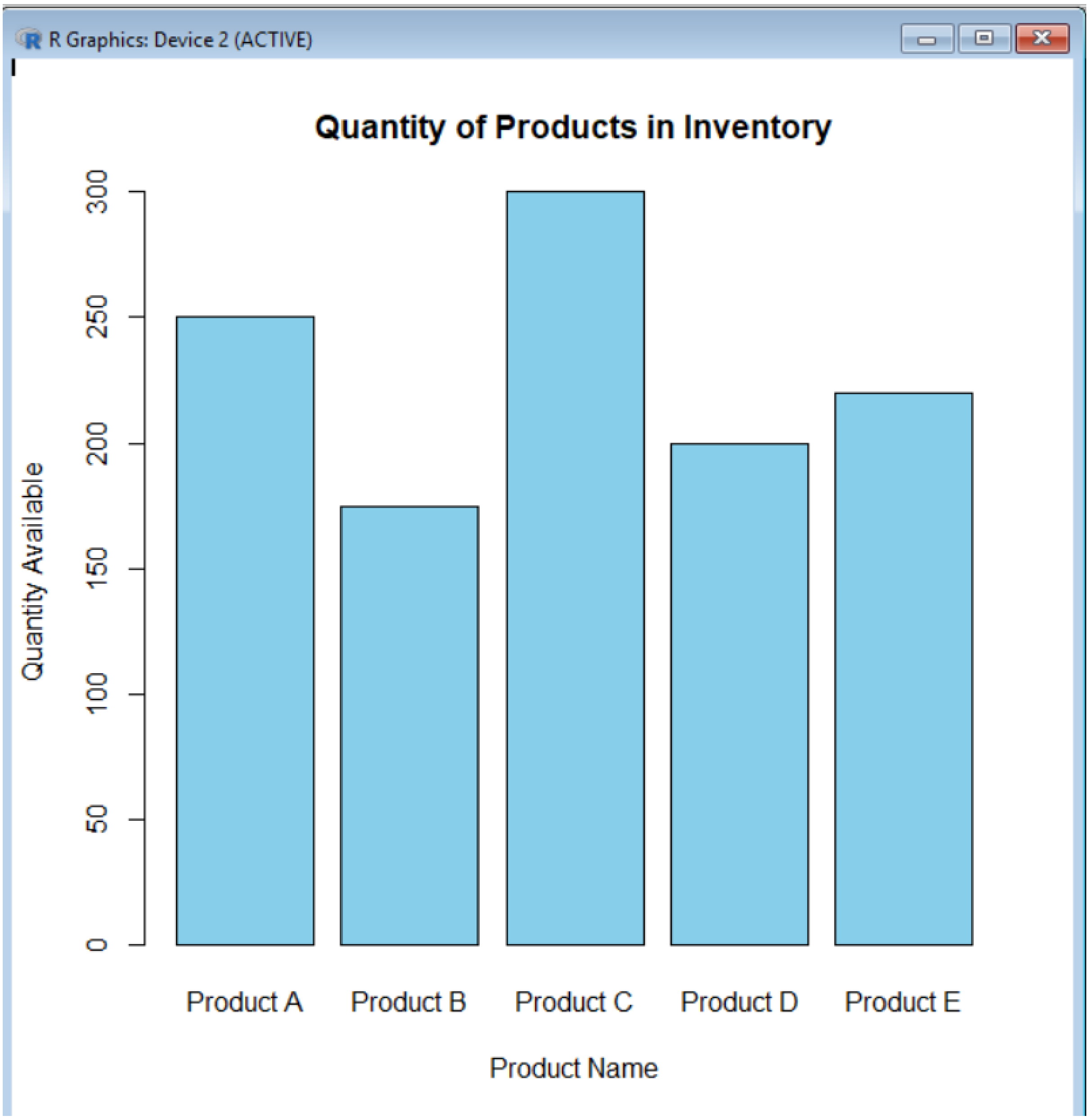

# Set 4

# Program 1: Bar Chart – Quantity of Each Product

```
ProductID <- c(1,2,3,4,5)
ProductName <- c("Product A","Product B","Product C","Product D","Product E")
QuantityAvailable <- c(250,175,300,200,220)
```

```
data <- data.frame(ProductID, ProductName, QuantityAvailable)
barplot(data$QuantityAvailable,
    names.arg=data$ProductName,
    col="skyblue",
    main="Quantity of Products in Inventory",
    xlab="Product Name",
    ylab="Quantity Available")
```

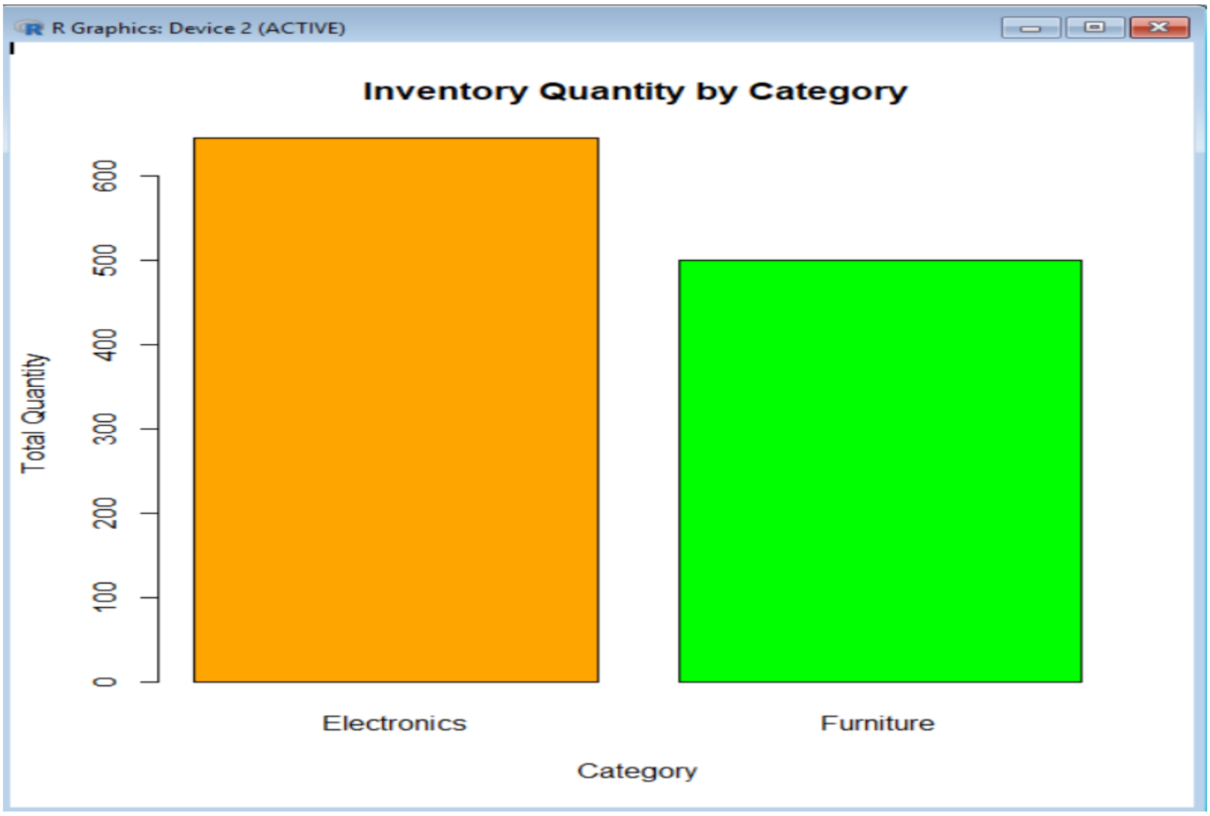
# Output:



# Program 2: Stacked Bar Chart – Product Quantity by Category

```
ProductID <- c(1,2,3,4,5)

ProductName <- c("Product A","Product B","Product C","Product D","Product E")

QuantityAvailable <- c(250,175,300,200,220)

Category <- c("Electronics","Electronics","Furniture","Furniture","Electronics")

data <- data.frame(ProductID, ProductName, QuantityAvailable, Category)

table_data <- tapply(data$QuantityAvailable,
                     data$Category,
                     sum)

barplot(table_data,
        col=c("orange","green"),
        main="Inventory Quantity by Category",
        xlab="Category",
        ylab="Total Quantity")
```

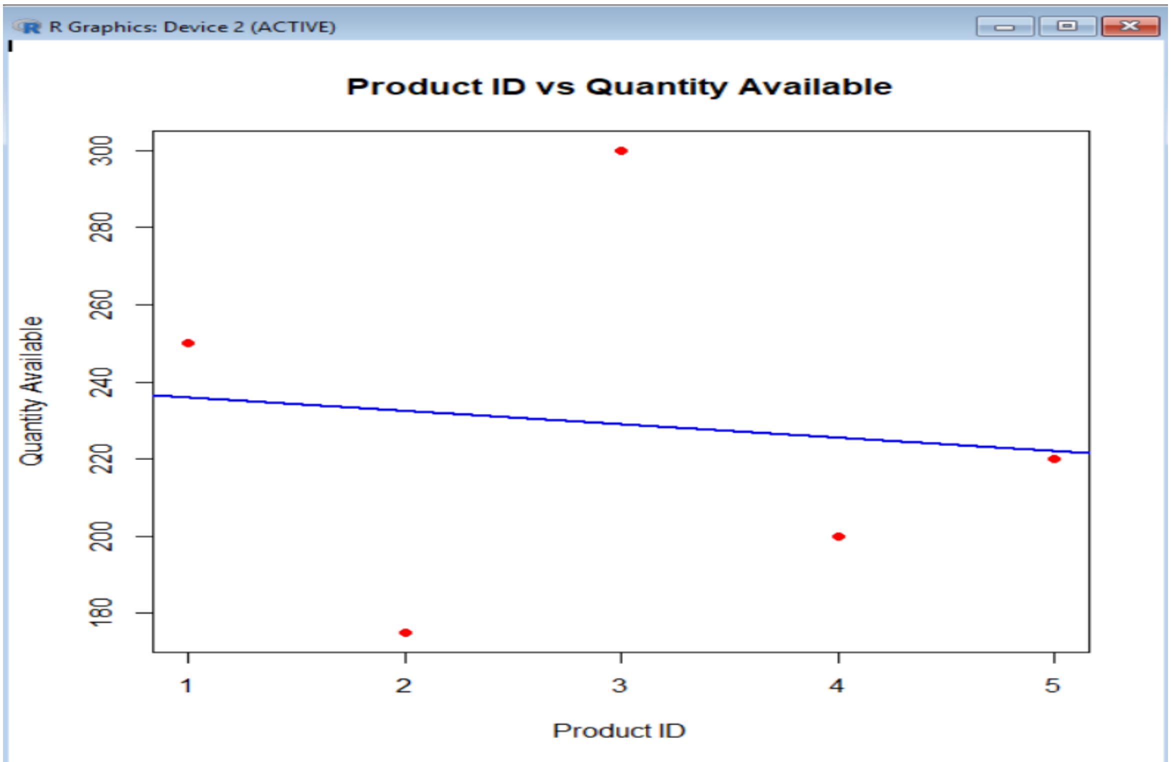
# Output:



# Program 3: Scatter Plot – Product ID vs Quantity Available

```
ProductID <- c(1,2,3,4,5)

ProductName <- c("Product A","Product B","Product C","Product D","Product E")

QuantityAvailable <- c(250,175,300,200,220)

data <- data.frame(ProductID, ProductName, QuantityAvailable)

plot(data$ProductID,
     data$QuantityAvailable,
     pch=19,
     col="red",
     xlab="Product ID",
     ylab="Quantity Available",
     main="Product ID vs Quantity Available")

abline(lm(QuantityAvailable ~ ProductID, data=data),
       col="blue",
       lwd=2)
```

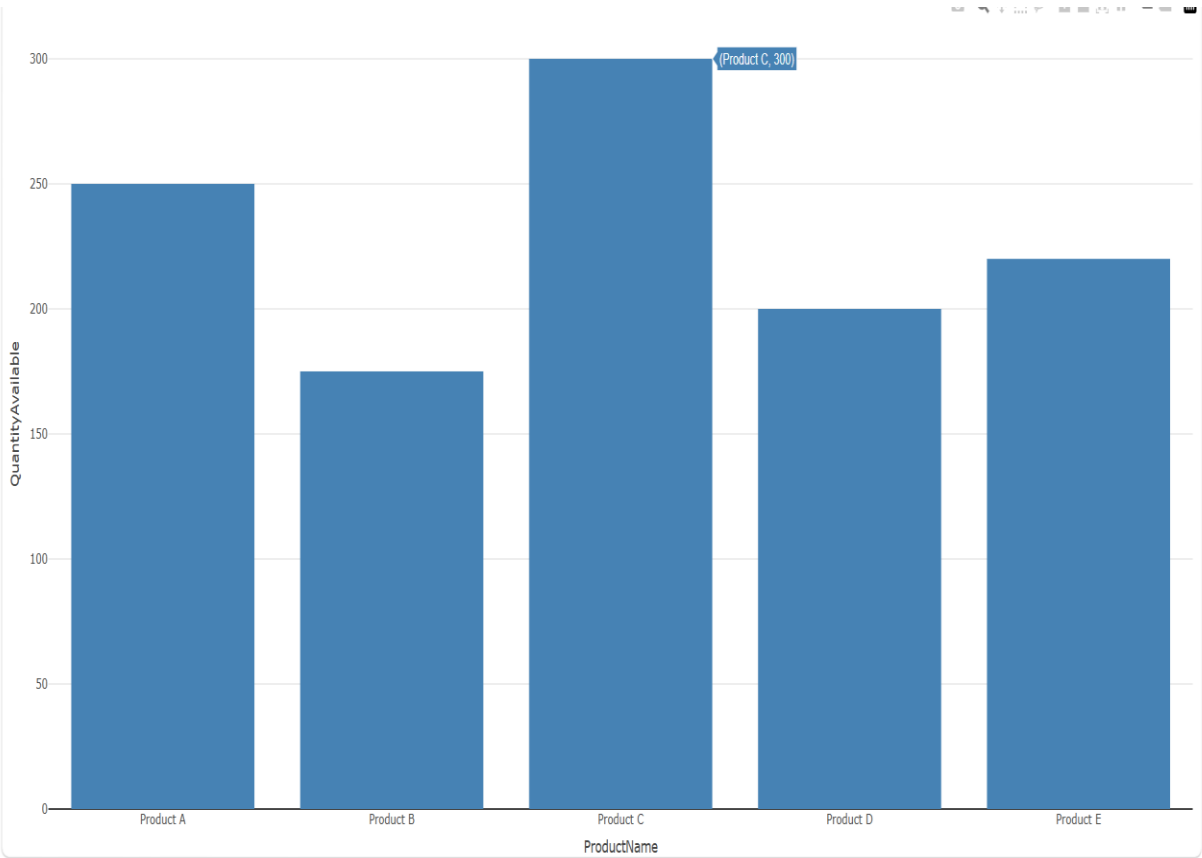
# Output:



# Program 4: Interactive Dashboard (Plotly)

```
library(plotly)
ProductID <- c(1,2,3,4,5)
ProductName <- c("Product A","Product B","Product C","Product D","Product E")
QuantityAvailable <- c(250,175,300,200,220)
data <- data.frame(ProductID, ProductName, QuantityAvailable)
fig <- plot_ly(data,
        x=~ProductName,
        y=~QuantityAvailable,
        type="bar",
        marker=list(color="steelblue"))
fig
```

# Output:

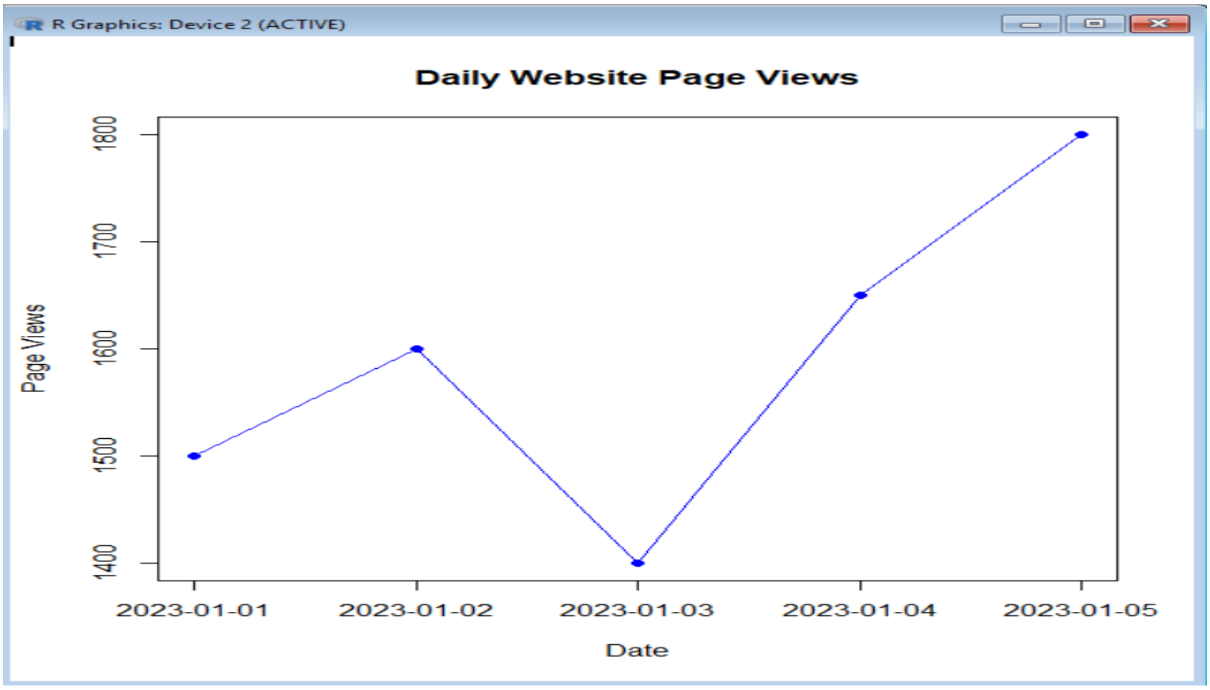


# Set 5

## Program 1: Line Chart – Daily Page Views Trend

```
Date <- c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")
PageViews <- c(1500,1600,1400,1650,1800)
CTR <- c(2.3,2.7,2.0,2.4,2.6)
data <- data.frame(Date, PageViews, CTR)
plot(PageViews,
    type="o",
    col="blue",
    pch=16,
    xaxt="n",
    xlab="Date",
    ylab="Page Views",
    main="Daily Website Page Views")
axis(1,
    at=1:5,
    labels=data$Date)
```

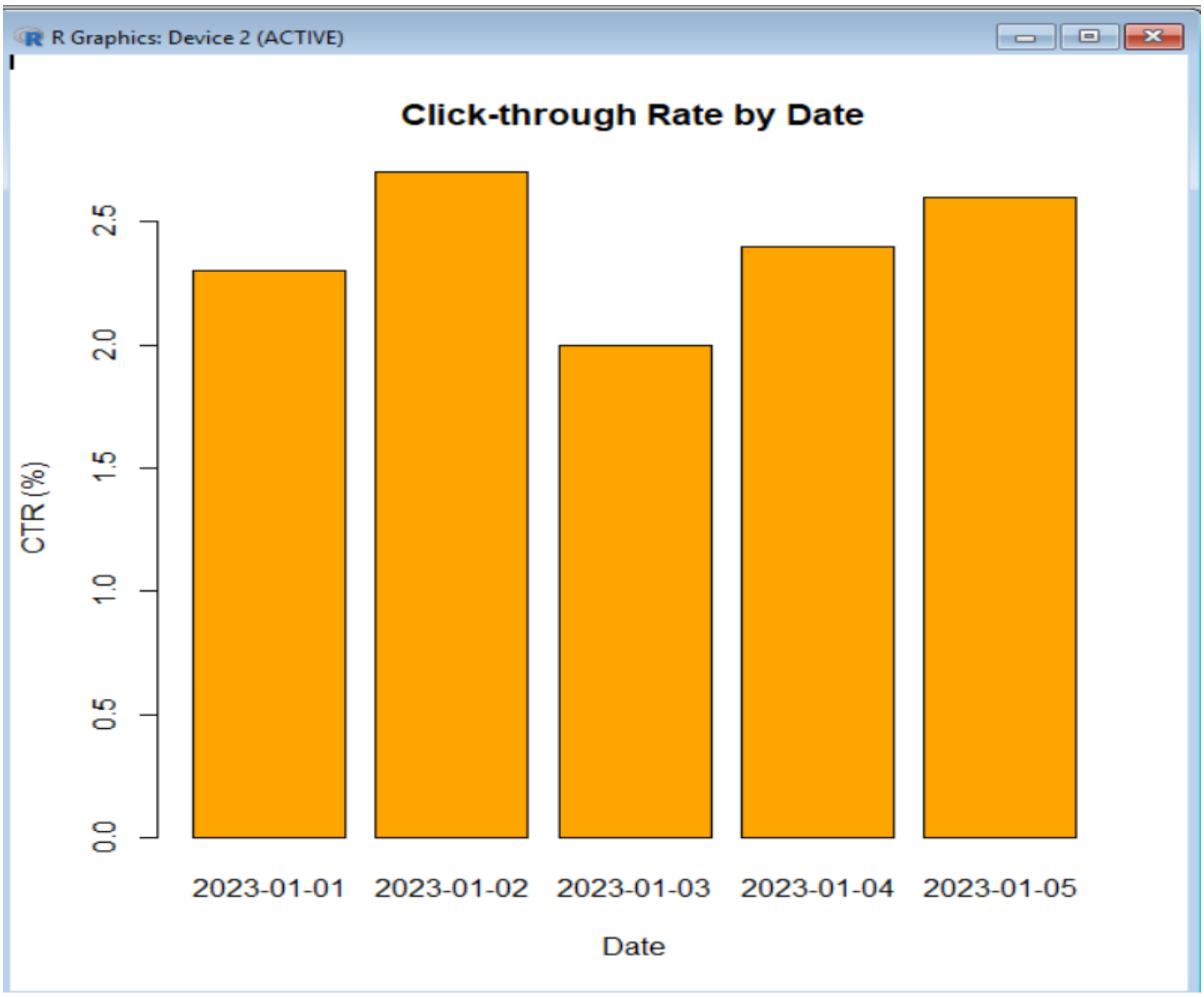
## Output:



# Program 2: Bar Chart – Click-through Rate (CTR)

```
Date <- c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")
PageViews <- c(1500,1600,1400,1650,1800)
CTR <- c(2.3,2.7,2.0,2.4,2.6)
data <- data.frame(Date, PageViews, CTR)
barplot(data$CTR,
        names.arg=data$Date,
        col="orange",
        main="Click-through Rate by Date",
        xlab="Date",
        ylab="CTR (%)")
```

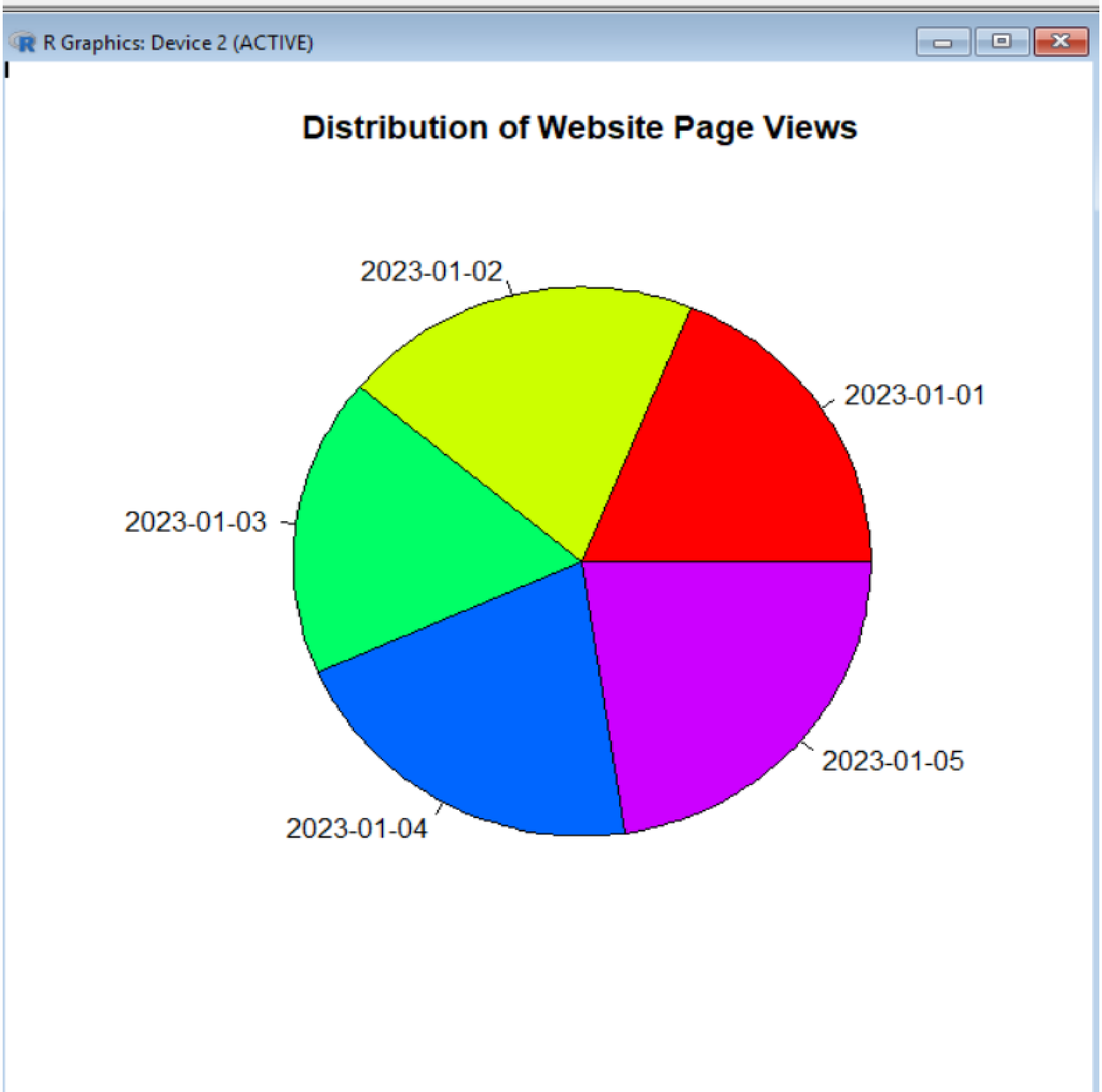
# Output:



## Program 3: Pie Chart – Distribution of Page Views

```
Date <- c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")
PageViews <- c(1500,1600,1400,1650,1800)
pie(PageViews,
   labels=Date,
   col=rainbow(5),
   main="Distribution of Website Page Views")
```

## Output:



# Program 4: Interactive Dashboard (Plotly)

```
library(plotly)

Date <- c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")

PageViews <- c(1500,1600,1400,1650,1800)

CTR <- c(2.3,2.7,2.0,2.4,2.6)

data <- data.frame(Date, PageViews, CTR)

fig <- plot_ly(data,
          x=~Date,
          y=~PageViews,
          type="scatter",
          mode="lines+markers",
          name="Page Views")

fig
```

# Output: